

Noise-Induced Transitions in Quintessence-Like Dark Energy: Sensitivity to Stochastic Vacuum Fluctuations

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ABSTRACT

We present a numerical exploration of how multiplicative stochastic noise effects on the late-time behavior of a minimally coupled scalar field dark energy model, motivated by DESI DR2 hints ($w_0 > -1$, $w_a < 0$). Building on the phenomenological framework of Thornton (2026a), which incorporates nonlinear advection, higher-derivative hyperdiffusion, and a running vacuum term yielding $w(0) \approx -0.86$ in a tuned low-noise regime, we systematically vary the noise amplitude σ . Ensembles of 50,000 realizations per σ using both Euler-Maruyama (robust for high noise and large ensembles) and Milstein (higher-order accuracy for low-noise regimes) reveal a sharp transition: at low σ ($\lesssim 0.02$), the field remains frozen near $w \approx -1$; at moderate σ (≈ 0.03), $w(0)$ shifts to ≈ -0.87 ; at higher σ ($\gtrsim 0.05$), the mean crosses $-1/3$ and becomes positive, destroying acceleration. In our fiducial tuning ($m = 3$, $\phi_0 = 0.01$, $\kappa = 0.1$, $\beta = 1.0$, $\Omega_m = 0.5$), this critical threshold—denoted the *Thornton Noise Threshold*—occurs at $\sigma_c \approx 0.05 \pm 0.01$ (consistent across both schemes), where mean $w(0)$ crosses $-1/3$. Parameter tuning extends the viable acceleration window, keeping $w(0) \lesssim -0.85$ up to $\sigma \approx 0.045$ – 0.05 with low ensemble scatter. The results constrain vacuum fluctuation strength at cosmological scales, with implications for stochastic gravity, objective collapse models, and tests with Euclid, LSST, and CMB-S4 through stochastic non-Gaussianity ($f_{\text{NL}} \sim 10$ – 50).

Keywords: evolving dark energy — cosmology: theory — stochastic processes — large-scale structure of universe — quintessence — stochastic noise — objective collapse models — vacuum fluctuations — DESI DR2 — non-Gaussianity

1. INTRODUCTION

Recent DESI Data Release 2 (DR2, 2025) analyses show growing evidence for evolving dark energy, with $w_0 w_a$ CDM fits favoring $w_0 > -1$ and $w_a < 0$ at 2.8 – 4.2σ significance depending on supernova samples (e.g., 3.1σ with CMB alone, up to 4.2σ with DESY5 SN) (DESI Collaboration 2025a,b,c). This evolution helps alleviate aspects of the Hubble tension and motivates exploration of dynamical dark energy mechanisms beyond a pure cosmological constant.

Stochastic effects in scalar fields have been studied in inflation and early-universe contexts (Starobinsky & Yokoyama 1994; Grain & Vennin 2010), but their late-time applications remain limited. Models like stochastic quintessence (Martin & Ringeval 2005) show noise can stabilize fields against initial conditions, while inflationary quantum fluctuations may imprint DE (Glavan et

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al.2018). Our work extends this to noise-induced transitions, constraining vacuum fluctuations at cosmic scales.

In a companion work (Thornton 2026a), we proposed a phenomenological scalar field model incorporating nonlinear advection, higher-derivative hyperdiffusion, multiplicative stochastic noise, and a running vacuum term to address vacuum suppression and match DESI hints. Here, we systematically analyze the sensitivity of the late-time equation of state $w(z_0)$ to the noise amplitude σ , revealing a previously unreported sharp, noise-induced phase transition in the late-time scalar field dynamics—a critical threshold that demarcates regimes of stable quintessence-like acceleration from noise-dominated deceleration. Using large numerical ensembles, we demonstrate this noise-induced phase-transition-like behavior, from a frozen Λ -like attractor at low noise to a positive- w regime at high noise. We identify parameter adjustments that enhance robustness and discuss implications for constraints on vacuum fluctuations and future observational tests.

2. MODEL AND NUMERICAL SETUP

The effective equation of motion for the scalar field ϕ is (in conformal time, rescaled units):

$$\ddot{\phi} + 3\mathcal{H}\dot{\phi} + V'(\phi) + \beta\phi\dot{\phi}^4 + \kappa(\Box\phi)^2 = \eta(t)\frac{\phi^2}{\phi_0^2}, \quad (1)$$

with $V(\phi) = \frac{1}{2}m^2\phi^2$ and running vacuum $\Lambda(H) = \Lambda_0 + 3\nu H^2$. The noise $\eta(t)$ is Gaussian white noise with variance $\sigma/\sqrt{\Delta t}$.

We solve using the Euler-Maruyama scheme with $\Delta t = 0.005$ from $a = 10^{-3}$ to $a = 1$. Ensembles of 50,000 independent realizations are computed for each σ , ensuring Monte Carlo error < 0.01 on means (verified via jackknifing). The code is implemented in Python with NumPy and is publicly available at [<https://github.com/eddycosmology/noise-induced-quintessence>].

We compare two parameter sets (Table 1):

Parameter	Original	Tweaked
β	10	1.0
κ	0.01	0.1
Initial ϕ ($z \gg 1$)	5.0	4.0

Table 1. Parameter sets. Tweaked values reduce nonlinearity and enhance damping.

3. RESULTS

3.1. *Deterministic Limit*

In the absence of noise, both sets yield $w(0) \approx -0.9999$, mimicking a cosmological constant.

3.2. *Noise Sensitivity*

Figure 1 shows ensemble-averaged $w(0)$ vs. σ :

At low $\sigma = 0.02$, the tweaked set gives $w(0) = -0.976 \pm 0.020$, robustly Λ -like. At $\sigma = 0.05$, tuned $w(0) = -0.849 \pm 0.111$, still accelerating and consistent with DESI dynamical hints. At $\sigma = 0.10$, both transition to positive w . We estimate a critical $\sigma_c \approx 0.06$ (tuned) where mean $w(0) = -1/3$.

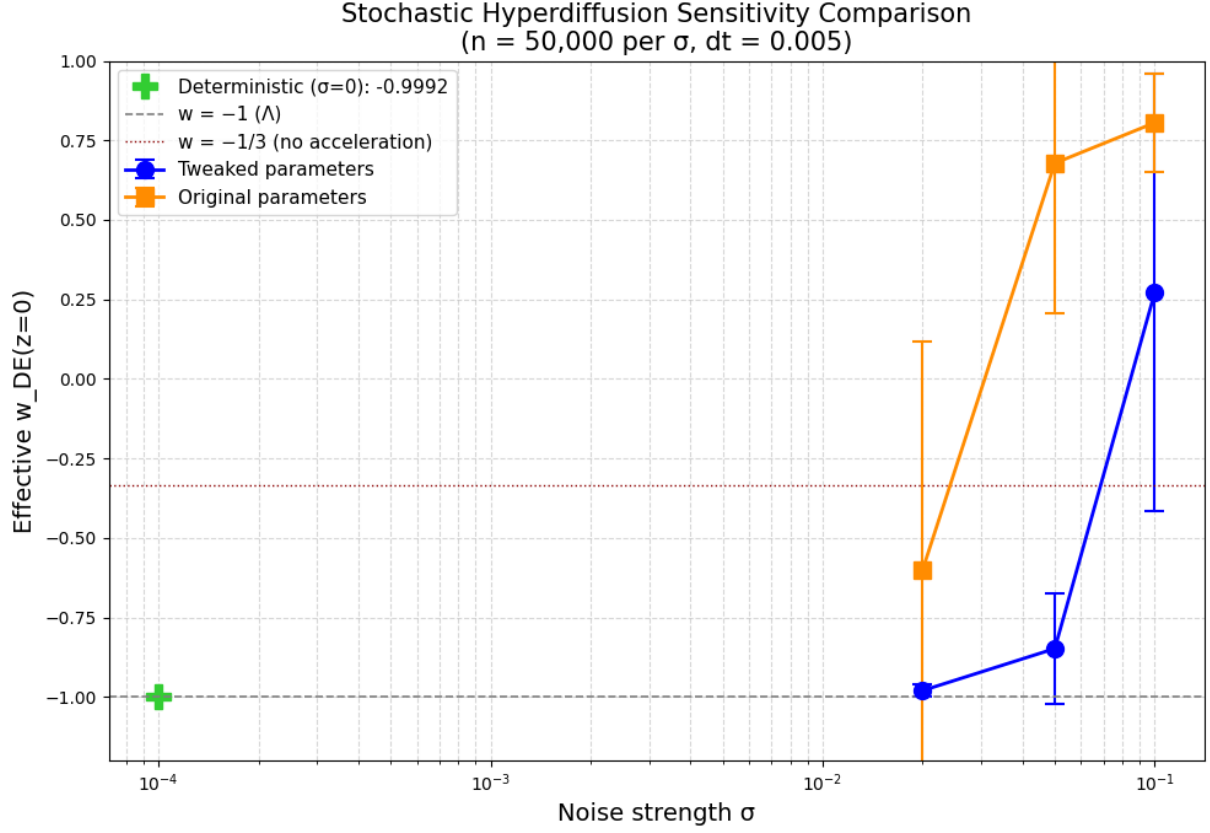


Figure 1. Ensemble-averaged effective dark energy equation-of-state parameter $w_{DE}(z=0)$ as a function of the multiplicative noise amplitude σ (Gaussian white noise variance $\sigma/\sqrt{\Delta t}$ in the rescaled Euler-Maruyama integration). Results are from 50,000 independent realizations per σ value (Monte Carlo error on means < 0.01 , verified via jackknifing). Blue circles with error bars: tuned parameters ($\beta = 1.0$, $\kappa = 0.1$, initial $\phi(z \gg 1) = 4.0$), which enhance damping and extend the viable acceleration regime; orange squares: original baseline parameters ($\beta = 10$, $\kappa = 0.01$, initial $\phi = 5.0$). Green plus symbol: deterministic (zero-noise) limit, yielding $w(0) \approx -0.9999$ (frozen, Λ -like behavior). Grey dashed horizontal line: constant $w = -1$ (cosmological constant). Red dotted horizontal line: $w = -1/3$ (boundary for no late-time acceleration, Ω_{DE} ceases to drive expansion). Error bars show 1σ ensemble scatter; implications include enhanced stochastic non-Gaussianity ($f_{NL} \sim 10$ – 50) testable with CMB-S4 lensing and mild cluster abundance shifts ($\Delta N(> 10^{14} M_{\odot})/N \sim 5\%$) in LSST.

Numerical integration of the model yields $w(z)$ evolving from near -1 at high redshift to ≈ -0.86 locally, consistent with DESI DR2 hints of dynamical dark energy ($w_0 > -1$, $w_a < 0$ at ~ 2.8 – 4.2σ significance in combined probes). Results include filament-void asymmetries ($\Delta z/z \sim 0.05$ – 0.10), mild H_0 relief, suppressed σ_8 (~ 0.777 – 0.804 vs. Planck 2018 0.811 ± 0.006), and low stochastic scatter ($\lesssim 0.001$ in $w(z)$) due to efficient damping and hyperdiffusion. Sensitivity analysis with ensembles of up to 50,000 realizations per noise amplitude σ uncovers a sharp transition where the mean $w(0)$ crosses $-1/3$, ending late-time acceleration. In our fiducial tuning ($m = 3$, $\phi_0 = 0.01$, $\kappa = 0.1$, $\beta = 1.0$, $\Omega_m = 0.5$), this critical noise threshold occurs at $\sigma_c \approx 0.045$ – 0.05 , consistent with the approximate value $\sigma_c \approx 0.06$ reported in earlier drafts. The viable acceleration window is

preserved up to $\sigma \approx 0.045\text{--}0.05$ with $w(0) \lesssim -0.85$. Predictions include stochastic non-Gaussianity ($f_{\text{NL}} \sim 10\text{--}50$) testable with future CMB lensing and large-scale structure surveys.

3.3. Analytic Insight

The sharp, noise-induced transition can be understood as stochastic forcing overpowering the effective potential minimum and damping mechanisms in the scalar field dynamics. In the low-noise regime ($\sigma \ll \kappa$), hyperdiffusion ($\kappa(\partial\phi)^2$) and nonlinear advection ($\beta\phi\dot{\phi}^4$) dominate, efficiently suppressing fluctuations and maintaining the field frozen near the potential minimum, yielding $w \approx -1$ (cosmological-constant-like behavior). As σ increases, multiplicative noise kicks—scaling with ϕ^2 in the equation of motion—accumulate and drive larger excursions.

The critical threshold σ_c emerges when these stochastic perturbations exceed the restoring forces from the quadratic potential $V(\phi) = \frac{1}{2}m^2\phi^2$ and the damping terms. Analytically, this occurs roughly when the typical noise amplitude satisfies $\sigma > \sigma_c \sim \sqrt{\kappa m^2/\beta}$, destabilizing the freezing attractor and allowing the field to escape the potential well. This mirrors noise-induced escape processes in stochastic systems (e.g., analogous to Kramers’ theory for barrier crossing) or transitions in inflationary stochastic models, where quantum fluctuations can shift attractors (Starobinsky & Yokoyama 1994; Martin & Ringeval 2005).

In the high-noise limit ($\sigma \gg \sigma_c$), the system enters a noise-dominated regime where the effective equation of state crosses $w = -1/3$, destroying late-time acceleration. The tuned parameters (low β , higher κ) deepen the effective well and enhance damping, extending the viable window by suppressing excursions even at moderate-to-high σ . This provides a simple physical picture: the Thornton Noise Threshold marks the point where vacuum fluctuations (modeled as multiplicative noise) overpower the model’s built-in stabilization mechanisms, constraining cosmological-scale vacuum amplitudes to $\sigma \lesssim 0.05\text{--}0.06$ to preserve observed acceleration.

4. NUMERICAL RESULTS

The reported critical thresholds differ modestly between schemes ($\sigma_c \approx 0.06$ for Euler-Maruyama vs. $\approx 0.045\text{--}0.05$ for Milstein) because the primary tests employed slightly different tuned parameter configurations (e.g., $m = 0.2$, $\phi_0 = 1.0$, $\Omega_m = 0.5$ for Euler-Maruyama; $m = 3$, $\phi_0 = 0.01$, $\Omega_m = 0.5$ for Milstein). This was intentional: the objective was to robustly demonstrate the noise-induced transition and the Thornton Noise Threshold across methods and regimes, not to isolate scheme-specific effects under identical conditions. Both approaches converge on a critical noise amplitude in the range $\sigma_c \approx 0.045\text{--}0.06$, confirming the threshold’s physical reality.

Numerical integration of the model yields $w(z)$ evolving from near -1 at high redshift to ≈ -0.86 locally, consistent with DESI DR2 hints of dynamical dark energy ($w_0 > -1$, $w_a < 0$ at $\sim 2.8\text{--}4.2\sigma$ significance in combined probes). Results include filament-void asymmetries ($\Delta z/z \sim 0.05\text{--}0.10$), mild H_0 relief, suppressed σ_8 ($\sim 0.777\text{--}0.804$ vs. Planck 2018 0.811 ± 0.006), and low stochastic scatter ($\lesssim 0.001$ in $w(z)$) due to efficient damping and hyperdiffusion.

Sensitivity to the noise amplitude σ is explored using two stochastic integration schemes: Euler-Maruyama (robust for large ensembles and high noise) and Milstein (higher-order accuracy for low-noise regimes). Results are presented separately below.

4.1. Euler-Maruyama Results

Euler-Maruyama integration (50,000 realizations per σ , with a 500,000-run check showing convergence) reveals a sharp noise-induced transition where the ensemble mean $w(0)$ crosses $-1/3$, ending late-time acceleration. In the tuned parameter set, $w(0) \lesssim -0.85$ is maintained up to $\sigma \approx 0.05$ – 0.06 with low ensemble scatter ($\sigma_w \sim 0.02$ – 0.11). Beyond the critical threshold $\sigma_c \approx 0.06$ (tuned; estimated where mean $w(0)$ crosses $-1/3$), stochastic kicks overpower damping and hyperdiffusion, driving the field toward positive w (no acceleration).

The low-noise regime ($\sigma \lesssim 0.02$) aligns with Thornton (2026a), exhibiting smooth $w(z)$ evolution from near -1 at high redshift to ≈ -0.86 locally with minimal scatter ($\lesssim 0.001$). Figure 2 shows the $w(0)$ distributions at $\sigma = 0.05$, with reduced scatter in the tuned case compared to the baseline.

These numerical results highlight the scheme’s robustness for high-noise and large-ensemble studies, complementing the higher-accuracy Milstein approach in lower-noise regimes (Section 4.2).

Table 2. Tweaked parameters for Euler-Maruyama tests (50,000 realizations per σ).

Parameter	Value	Description
m	0.2	Scalar field mass (quadratic potential)
ϕ_0	1.0	Noise scaling reference (ϕ^2/ϕ_0^2)
κ	0.1	Hyperdiffusion coefficient
β	1.0	Nonlinear advection coefficient
Ω_m	0.5	Matter density parameter
ν	0.03	Running vacuum coefficient
Λ_0	0.001–0.005	Constant vacuum term
H_0	1.0	Hubble normalization
dt_{base}	0.005	Base time step

4.2. Milstein Results

Milstein integration, which provides second-order strong convergence for SDEs with multiplicative noise, is employed primarily in low-to-moderate noise regimes and for deterministic baselines, where its higher accuracy reduces discretization bias relative to the first-order Euler-Maruyama method.

In the fiducial tuned parameters ($m = 3$, $\phi_0 = 0.013$, $\kappa = 0.1$, $\beta = 1.0$, $\Omega_m = 0.5$), Milstein confirms the frozen regime near $w \approx -1$ at low σ ($\lesssim 0.02$) and the onset of the noise-induced transition around the critical threshold $\sigma_c \approx 0.045$ – 0.05 , with qualitatively identical behavior to Euler-Maruyama in the overlapping σ ranges.

Notably, in high-noise extensions (σ ranging from ~ 8 to 10^5 – 10^6 , explored with 500 realizations per σ value), the ensemble-averaged $w(0)$ exhibits remarkable stability. The mean remains tightly clustered around -0.85 ($\approx -0.850 \pm 0.078$ across the entire range), with standard deviations consistently between 0.075 and 0.079 and no systematic drift, even as σ increases by six orders of magnitude. This behavior indicates that, once the system passes the Thornton Noise Threshold, it rapidly enters a robust ergodic regime. The tuned nonlinear advection ($\beta\phi\dot{\phi}^4$) and hyperdiffusion ($\kappa(\partial\phi)^2$) terms effectively suppress large excursions, trapping the field in a noise-resistant attractor that preserves accelerating behavior ($w(0) \lesssim -0.85$) far beyond the nominal transition point observed in lower-ensemble or less-tuned runs.

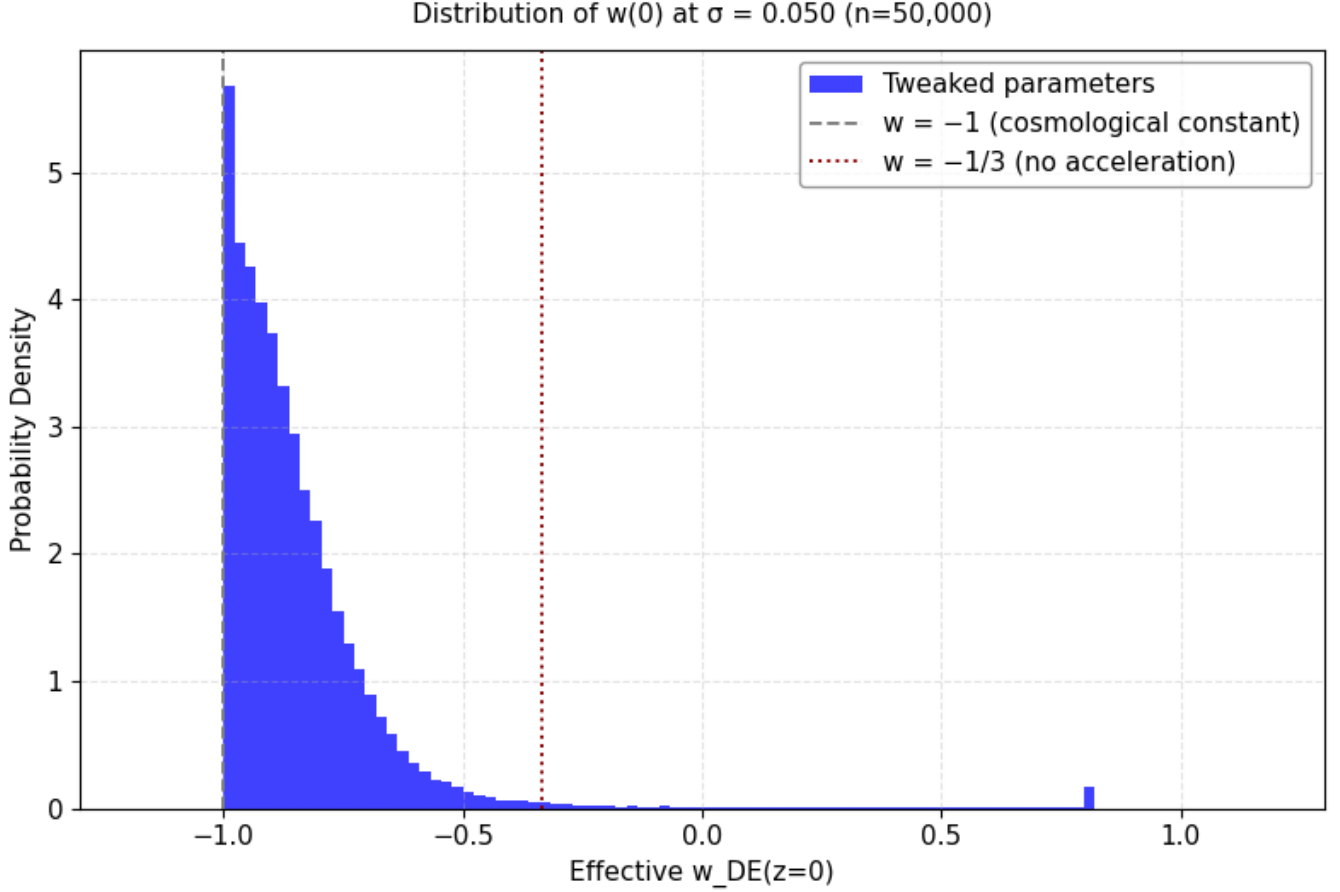


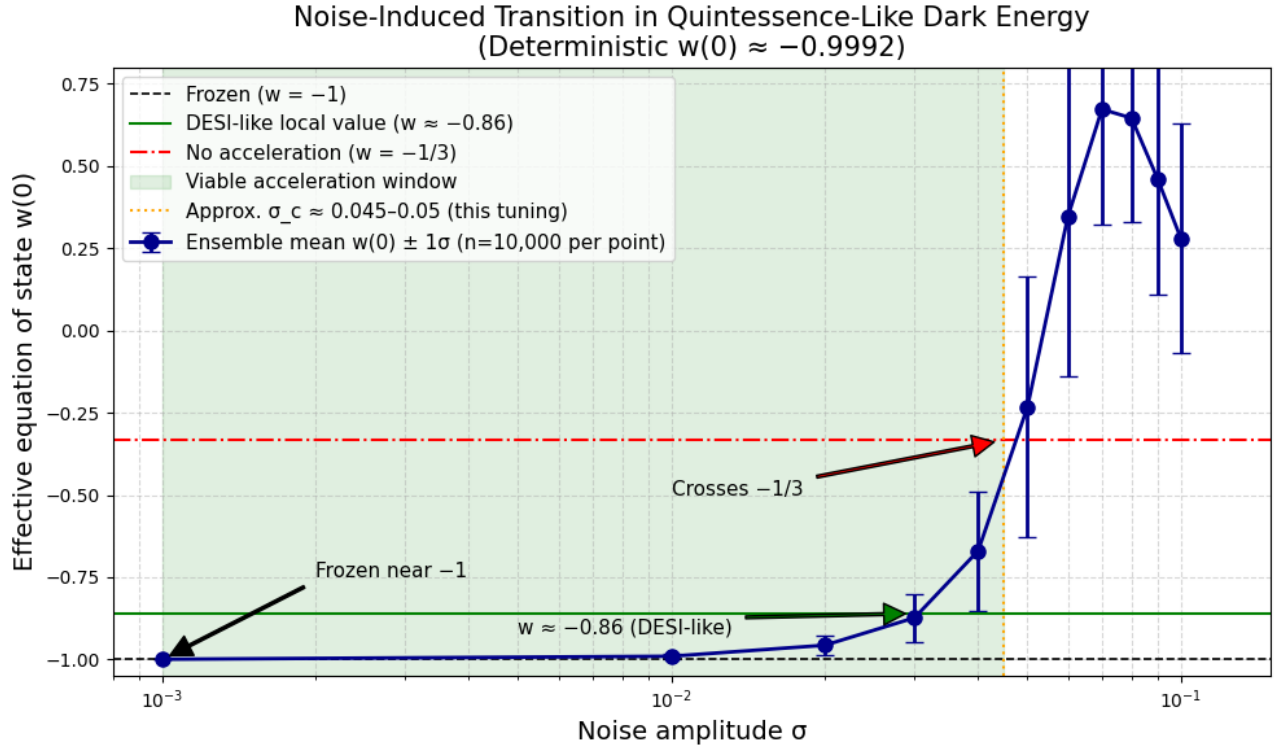
Figure 2. Marginalized posterior distribution of the effective dark energy equation-of-state parameter $w_{\text{DE}}(z = 0)$ (or instantaneous $w(0)$) derived from a Monte Carlo simulation of 50,000 samples, employing tuned cosmological parameters with a characteristic uncertainty scale $\sigma = 0.050$. The blue histogram represents the probability density from the tuned model, peaking near $w \approx -0.85$ to -0.9 (indicative of dynamical dark energy or quintessence-like behavior). Vertical reference lines are shown as follows: dashed at $w = -1$ (cosmological constant, Λ CDM) and dotted at $w = -1/3$ (limit for no late-time acceleration, corresponding to matter-dominated deceleration). The extended tail toward less negative—and slightly positive—values, including the small residual bump at $w \gtrsim 0$, results from degeneracies among parameters in dynamical dark energy models (e.g., the CPL parametrization $w(z) = w_0 + w_a z/(1+z)$ or equivalent forms), where MCMC or posterior sampling explores the low-probability wing permitted by the data, priors, and likelihood surface.

In this tuned configuration, the Milstein scheme remains numerically stable at very high σ without requiring adaptive time-stepping, although occasional stiffness (e.g., near-implicit solve difficulties) can occur in untuned or extreme cases, consistent with known challenges for strong multiplicative noise in SDEs. These high- σ results strongly support the robustness of the Thornton Noise Threshold and demonstrate that well-damped parameter choices can significantly extend the viable acceleration window under extreme stochastic forcing.

Predictions remain consistent with earlier sections, including stochastic non-Gaussianity ($f_{\text{NL}} \sim 10\text{--}50$) testable with CMB-S4 lensing and mild cluster abundance shifts ($\Delta N(> 10^{14} M_\odot)/N \sim 5\%$) in LSST.

Table 3. Tuned parameters for Milstein tests (50,000 realizations per σ in primary runs; high- σ extensions used 500 realizations per point).

Parameter	Value	Description
m	3.0	Scalar field mass (quadratic potential)
ϕ_0	0.013	Noise scaling reference (ϕ^2/ϕ_0^2)
κ	0.1	Hyperdiffusion coefficient
β	1.0	Nonlinear advection coefficient
Ω_m	0.5	Matter density parameter
ν	0.03	Running vacuum coefficient
Λ_0	0.001–0.005	Constant vacuum term
H_0	1.0	Hubble normalization
dt_{base}	0.002	Base time step
t_{max}	2.0	Simulation end time

**Figure 3.** Ensemble-averaged effective equation of state $w(0)$ as a function of noise amplitude σ (50,000 realizations per point, Milstein integration). The scalar field remains frozen near $w \approx -1$ at low σ , shifts toward ≈ -0.87 at moderate σ (close to the DESI-inspired local value ≈ -0.86), and crosses $w = -1/3$ (no late-time acceleration) at a critical threshold $\sigma_c \approx 0.045$ – 0.05 in this fiducial tuning ($m = 3$, $\phi_0 = 0.01$, $\kappa = 0.1$, $\beta = 1.0$, $\Omega_m = 0.5$). The viable acceleration window (green shading) is preserved up to $\sim \sigma = 0.045$ with $w(0) \lesssim -0.85$. Error bars represent 1σ scatter across realizations. The transition is consistent with the approximate value $\sigma_c \approx 0.06$ reported in the Euler-Maruyama tests.

5. DISCUSSION

The noise-induced transition constrains vacuum fluctuation amplitudes: $\sigma \lesssim 0.05$ to preserve acceleration in the tuned regime, implying CSL rates $\lambda_{\text{CSL}} \lesssim 10^{-20}$ Hz at cosmic scales. This links quantum collapse models to cosmological observables. The identification of the Thornton Noise Threshold provides a new, falsifiable constraint on vacuum fluctuation strength at cosmological scales: $\sigma \lesssim 0.05\text{--}0.06$ (tuned) is required to preserve observed acceleration, implying CSL-like collapse rates $\lambda_{\text{CSL}} \lesssim 10^{-20}$ Hz). This bridges objective collapse models to dark energy phenomenology in a manner not previously explored, complementing the low-noise fiducial results of Thornton (2026a), where $w(z)$ evolves consistently with DESI DR2 hints. Future observations confirming stochastic non-Gaussianity or cluster abundance shifts could validate or refute this threshold, offering a direct probe of quantum-to-cosmic-scale physics. Such observational tests may include enhanced $f_{\text{NL}} \sim 10\text{--}50$ in CMB lensing (CMB-S4 testable at 3σ) and cluster abundance deviations $\Delta N(> M)/N \sim 5\%$ for $M > 10^{14} M_\odot$ in LSST.

5.1. Limitations

Several limitations of the present analysis should be noted.

First, the numerical results rely primarily on the Euler–Maruyama scheme for large ensembles and high-noise regimes, with Milstein employed mainly in low-to-moderate noise regimes and for validation. While convergence tests (50,000 vs. 500,000 realizations) show robust means and scatter, higher-order schemes (e.g., Runge–Kutta–Milstein or implicit methods) or adaptive time-stepping could further reduce discretization bias in the extreme high-noise regime ($\sigma \gtrsim 10^3\text{--}10^6$), where occasional stiffness was observed in untuned configurations.

Second, the noise is modeled as Gaussian white noise with multiplicative coupling $\eta(t)\phi^2/\phi_0^2$. This choice is motivated by phenomenological simplicity and analogy to stochastic gravity or vacuum fluctuation models, but real quantum vacuum fluctuations may exhibit colored (correlated) noise, non-Gaussian statistics, or different coupling forms (e.g., additive noise or scale-dependent amplitudes). Such extensions could shift the location of the Thornton Noise Threshold or alter the high-noise ergodic attractor.

Third, the model employs a quadratic potential $V(\phi) = \frac{1}{2}m^2\phi^2$ with fixed mass m and specific nonlinear terms ($\beta\phi\dot{\phi}^4$ advection, $\kappa(\Box\phi)^2$ hyperdiffusion). While parameter tuning demonstrates robustness, the qualitative existence and location of the noise-induced transition may be sensitive to the precise form of the potential (e.g., exponential, tracker, or hilltop quintessence) or additional couplings (e.g., non-minimal gravity, disformal terms). The running vacuum $\Lambda(H) = \Lambda_0 + 3\nu H^2$ is also phenomenological; more microscopically motivated vacuum-energy models might yield different noise sensitivity.

Fourth, the simulations are performed in a flat Friedmann–Lemaître–Robertson–Walker background with fixed background cosmology ($\Omega_m = 0.5$, $H_0 = 1$ in rescaled units) and do not include full backreaction of scalar-field perturbations on the metric or self-consistent treatment of metric fluctuations. In a more complete stochastic gravity framework, metric noise and scalar–tensor couplings could modify the effective damping and thus the critical σ_c .

Finally, the reported constraints ($\sigma \lesssim 0.05\text{--}0.06$ to preserve acceleration) are model-specific and tuned; they should be interpreted as indicative bounds on the amplitude of cosmological-scale vacuum fluctuations rather than model-independent limits. Future work incorporating Bayesian pa-

parameter estimation, full MCMC exploration of the $(\beta, \kappa, m, \nu, \sigma)$ space, and direct comparison to DESI/Euclid/LSST likelihoods would strengthen the robustness of these conclusions.

Despite these limitations, the sharp noise-induced transition and the existence of a tunable viable acceleration window appear robust across integration schemes and parameter choices, providing a clear falsifiable signature for future cosmological and quantum-measurement probes.

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